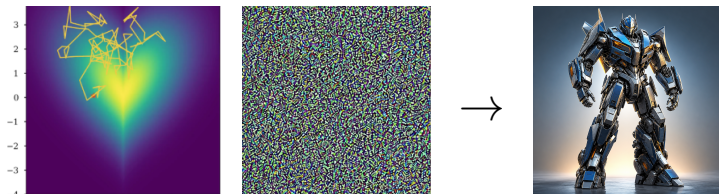


On Regularization of Maximum Mean Discrepancy Gradient Flow

Background

Problem: How to learn a target probability distribution π on $\mathbb{R}^d$.

- Sampling (e.g: $\pi \propto \exp(-V)$ is the posterior distribution in Bayesian inference).
- Generative models (e.g: π is the distribution of an image dataset).



Background: Optimization in the space of probability measures

Problem: How to learn a target probability distribution $\mathbb{Q}$ on $\mathbb{R}^d$.

- This problem can be written as an **optimization** problem on $\mathcal{P}_2(\mathbb{R}^d)$.

$$\arg \min_{\mathbb{P} \in \mathcal{P}_2(\mathbb{R}^d)} D(\mathbb{P} \mid \mathbb{Q}).$$

- Here D is a similarity metric or distance, e.g. Kullback–Leibler divergence.
 - $D(\mathbb{P} \mid \mathbb{Q}) = 0$ if and only if $\mathbb{P} = \mathbb{Q}$.
- $\mathcal{P}_2(\mathbb{R}^d)$ denotes the space of probability measures with a finite second moment.
 - We primarily consider $\mathcal{P}_2(\mathbb{R}^d)$ rather than $\mathcal{P}(\mathbb{R}^d)$ for the nice geometrical properties.
- How to find the minimum? Gradient descent!

Background: Euclidean gradient flow

- Euclidean gradient flow of an objective $F : \mathbb{R}^d \rightarrow \mathbb{R}$

$$\partial_t x_t = -v(x_t), \quad v = \nabla F.$$

- ∇F denotes the gradient of F .
- This is the continuous-time analogue of gradient descent:

$$x_{n+1} = x_n - \gamma v(x_n),$$

where $\gamma > 0$ is the step size.

- When F is both **strongly convex and smooth**, Euclidean gradient **descent** converges exponentially fast [BV04].

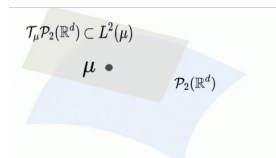
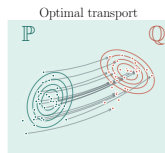
Background: Wasserstein gradient flow

Challenge: How to find $\arg \min_{\mathbb{P} \in \mathcal{P}_2(\mathbb{R}^d)} D(\mathbb{P} \mid \mathbb{Q})$?

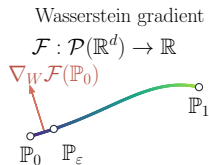
- Gradient descent! Wait... How do we define gradients in $\mathcal{P}_2(\mathbb{R}^d)$?
- Endow $\mathcal{P}_2(\mathbb{R}^d)$ with the Wasserstein-2 distance W_2 .

$$W_2^2(\nu, \mathbb{P}) = \int \|T(x) - x\|^2 d\nu(x) = \|T - \text{Id}\|_{L_2(\nu)}^2.$$

- $W_2^2(\nu, \mathbb{P})$ means the **minimal energy** takes to transport mass from ν to $\mathbb{P}$.
- $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is the **optimal transport map** from ν to $\mathbb{P}$.
- $(\mathcal{P}_2, W_2)$ can be 'treated' as a Riemann manifold under Otto's calculus [Ott01].
- The **tangent space** $\mathcal{T}_{\mathbb{P}}\mathcal{P}_2(\mathbb{R}^d)$ at $\mathbb{P} \in \mathcal{P}_2(\mathbb{R}^d)$ is a dense subset of $L_2(\mathbb{P})$.



Background: Wasserstein gradient flow



Definition (Wasserstein gradient)

Let $\mathcal{F} : \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathbb{R}$. The **Wasserstein gradient of $\mathcal{F}$** evaluated at $\mathbb{P} \in \mathcal{P}_2(\mathbb{R}^d)$ is the unique function $\nabla_{W_2} \mathcal{F}(\mathbb{P}) : \mathbb{R}^d \rightarrow \mathbb{R}^d$, s.t. for any $T \in \mathcal{T}_{\mathbb{P}} \mathcal{P}_2(\mathbb{R}^d)$,

$$\lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} [\mathcal{F}((\text{Id} + \epsilon T)_{\#} \mathbb{P}) - \mathcal{F}(\mathbb{P})] = \langle \nabla_{W_2} \mathcal{F}, T \rangle_{L_2(\mathbb{P})}.$$

- The gradient is defined along a ‘curve’ $(\text{Id} + \epsilon T)_{\#} \mathbb{P}$ in $\mathcal{P}_2(\mathbb{R}^d)$.

Background: Wasserstein gradient flow

Definition (Wasserstein gradient flow)

Wasserstein gradient flow of $\mathcal{F} : \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathbb{R}$ is a sequence of distributions $(\mathbb{P}_t)_{t \geq 0}$ that evolves according to the **continuity equation** $\partial_t \mathbb{P}_t + \nabla \cdot (\mathbb{P}_t v_t) = 0$, where the vector field $v_t = -\nabla_{W_2} \mathcal{F}(\mathbb{P}_t)$. Or equivalently, $\dot{x}_t = v_t(x_t)$ with $\mathbb{P}_t = \text{Law}(x_t)$.

Euclidean Gradient Flow

- State space: $\mathbb{R}^d$
- Objective $F : \mathbb{R}^d \rightarrow \mathbb{R}$.
- Update scheme: $\dot{x}_t = v_t(x_t)$, with $v_t = -\nabla F$.

Wasserstein Gradient Flow

- State space: $\mathcal{P}_2(\mathbb{R}^d)$
- Objective $\mathcal{F} : \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathbb{R}$:
- Update scheme $\dot{x}_t = v_t(x_t)$, with $v_t = -\nabla_{W_2} \mathcal{F}(\mathbb{P}_t)$.

Example 1: Langevin diffusion [JKO98]

- Given the target distribution $\mathbb{Q} \propto \exp(-V)$ with $V : \mathbb{R}^d \rightarrow \mathbb{R}$.
- The functional $\mathcal{F}_{\text{KL}} = \text{KL}(\cdot, \mathbb{Q})$ and its Wasserstein gradient

$$[\nabla_{W_2} \mathcal{F}_{\text{KL}}(\mathbb{P})](\cdot) = \nabla V(\cdot) + \nabla \log \mathbb{P}_t(\cdot).$$

- The Wasserstein gradient flow of $\mathcal{F}_{\text{KL}}$

$$\partial_t \mathbb{P}_t = \nabla \cdot (\mathbb{P}_t (\nabla V + \nabla \log \mathbb{P}_t)).$$

- It is equivalent to the Fokker–Planck equation of the **Langevin diffusion** [SS19]:

$$dx_t = -\nabla V(x_t) dt + \sqrt{2} dW_t, \quad \mathbb{P}_t = \text{Law}(x_t).$$

- A standard time-discretization (Euler–Maruyama scheme) is:

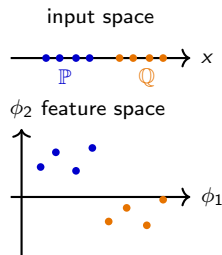
$$x_{n+1} = x_n - \gamma \nabla V(x_n) + \sqrt{2\gamma} \xi_n, \quad \xi_n \sim \mathcal{N}(0, \text{Id}_d).$$

Background: Maximum Mean Discrepancy (MMD) [GBR⁺12]

- $\mathcal{H}$ is a reproducing kernel Hilbert space (RKHS) with kernel k .

$$\text{MMD}(\mathbb{P}, \mathbb{Q}) := \sup_{f \in \mathcal{H}} |\mathbb{E}_{\mathbb{P}}[f(X)] - \mathbb{E}_{\mathbb{Q}}[f(X)]|.$$

- The RKHS contains functions that are linear combinations of the feature maps $\phi : \mathcal{X} \rightarrow \mathcal{H}$, $f = \sum_{i=1}^n \alpha_i \phi(x_i)$.



Theorem 1

MMD admits a closed-form expression

$$\text{MMD}^2(\mathbb{P}, \mathbb{Q}) = \mathbb{E}_{X, X' \sim \mathbb{P}}[k(X, X')] + \mathbb{E}_{Y, Y' \sim \mathbb{Q}}[k(Y, Y')] - 2\mathbb{E}_{X \sim \mathbb{P}, Y \sim \mathbb{Q}}[k(X, Y)].$$

- MMD is a probability divergence ($\text{MMD}(\mathbb{P}, \mathbb{Q}) = 0 \iff \mathbb{P} = \mathbb{Q}$) if k is characteristic (Gaussian) [SFL11].

Example 2: MMD gradient flow [AKSG19]

- The functional $\mathcal{F}_{\text{MMD}} = \frac{1}{2}\text{MMD}^2(\cdot, \mathbb{Q})$
- Its Wasserstein gradient

$$[\nabla_{W_2} \mathcal{F}_{\text{MMD}}(\mathbb{P})](\cdot) = \nabla \left(\int k(x, \cdot) d\mathbb{P}(x) - \int k(x, \cdot) d\mathbb{Q}(x) \right).$$

- The Wasserstein gradient flow of $\mathcal{F}_{\text{MMD}}$,

$$\partial_t \mathbb{P}_t = \nabla \cdot (\mathbb{P}_t \nabla_{W_2} \mathcal{F}_{\text{MMD}}(\mathbb{P}_t)), \quad dx_t = -[\nabla_{W_2} \mathcal{F}_{\text{MMD}}(\mathbb{P}_t)](x_t) dt.$$

- A standard time-discretization (Euler–Maruyama scheme) is:

$$x_{n+1} = x_n - \gamma \nabla_{W_2} \mathcal{F}_{\text{MMD}}(\mathbb{P}_n)(x_n), \quad \text{Law}(x_n) = \mathbb{P}_n.$$

- The same holds for $\mathcal{F}_{\text{KSD}} = \frac{1}{2}\text{MMD}^2(\cdot, \mathbb{Q})$ with a Stein kernel $k_{\mathbb{Q}}$ [KAFMA21].

Wasserstein gradient flow of $\mathcal{F}_{\text{KL}}$

- $\mathbb{Q} \propto \exp(-V)$
- Sampling
- Convergence in discrete time

$$\text{KL}(\mathbb{P}_n, \mathbb{Q}) \leq \exp(-\beta^{-1}\gamma n) \text{KL}(\mathbb{P}_0, \mathbb{Q}) + \gamma n L^2 \beta.$$

- β^{-1} is the LSI constant of $\mathbb{Q}$, and L is the Lipschitz constant of ∇V .
- $\lim_{n \rightarrow \infty} \text{KL}(\mathbb{P}_n, \mathbb{Q}) = 0!$.

Wasserstein gradient flow of $\mathcal{F}_{\text{MMD}}$

- $\{y_i\}_{i=1}^M$ i.i.d samples from $\mathbb{Q}$
- Generative modelling
- Non-convergence in discrete time

$$\text{MMD}^2(\mathbb{P}_n, \mathbb{Q}) \leq \frac{W_2^2(\mathbb{P}_0, \mathbb{Q})}{\gamma n} + \bar{K}.$$

- $\bar{K}$ is a positive barrier term that does not vanish.
- $\lim_{n \rightarrow \infty} \text{MMD}(\mathbb{P}_n, \mathbb{Q}) \neq 0!$

Non-convexity of MMD prevents global convergence

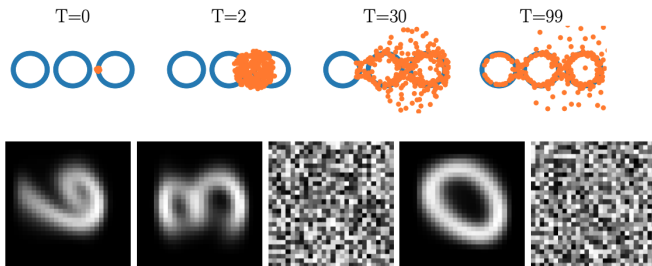


Figure: [BSM25]

- Convergence of MMD flow has been proved under noise injection and stringent conditions on the noise scale [AKSG19, CKK⁺25].
- Disclaimer: There exists many papers where MMD flow **empirically** generate high-quality images with adversarial training of kernels [GBG25], or with non-smooth kernels plus deep network distillation [HWAH24, AHS23].

Non-convexity of MMD prevents global convergence

Question: How can we overcome the non-convexity of $\mathcal{F}_{\text{MMD}}$?

- In this talk, I will present three complementary routes.

1. De-regularized MMD gradient flow

arXiv:2409.14980

2. Sobolev-regularized MMD gradient flow

arXiv:2605.11884

3. Applications: 1) Statistical inference; 2) Generative model

arXiv:2607.03871, *Work in progress*

DrMMD flow and SrMMD flow



Zonghao
Chen



Aratrika
Mustafi



Pierre
Glaser



Anna
Korba



Arthur
Gretton



Bharath K.
Sriperumbudur

Kernel assumptions

Assumption 1 (Growth I)

There exists a constant $M > 0$ such that, for every $x \in \mathbb{R}^d$,

$$\begin{aligned} |k(x, x)| &\leq M (1 + |x|^4), \\ |\partial_{1,i} \partial_{2,i} k(x, x)| &\leq M (1 + |x|^2), \quad i \in [d]. \end{aligned}$$

Assumption 2 (Local Boundedness)

For $R > 0$, let $\mathcal{K} = B(0, R) \subset \mathbb{R}^d$. There exists a $B_R > 0$ such that, for every $i, j \in [d]$,

$$\sup_{x \in \mathcal{K}} |\partial_{1,i} \partial_{1,j} \partial_{2,i} \partial_{2,j} k(x, x)| \leq B_R.$$

- Gaussian, Matérn, and inverse multiquadratic kernels.
- Polynomial kernels.
- Stein kernels with score functions of at most linear growth.

$T_{\mathbb{Q}} : L_2(\mathbb{Q}) \rightarrow L_2(\mathbb{Q})$ is the kernel **integral operator** defined as

$$T_{\mathbb{Q}}f(\cdot) = \int k(x, \cdot) f(x) d\mathbb{Q}(x).$$

Proposition 1 (MMD and χ^2 -divergence)

Suppose $\mathbb{P}$ is absolutely continuous with respect to $\mathbb{Q}$, i.e., $\mathbb{P} \ll \mathbb{Q}$. Then

$$\text{MMD}^2(\mathbb{P}, \mathbb{Q}) = \left\| T_{\mathbb{Q}}^{\frac{1}{2}} \left(\frac{d\mathbb{P}}{d\mathbb{Q}} - 1 \right) \right\|_{L_2(\mathbb{Q})}^2 \quad \text{and} \quad \chi^2(\mathbb{P}, \mathbb{Q}) = \left\| \frac{d\mathbb{P}}{d\mathbb{Q}} - 1 \right\|_{L_2(\mathbb{Q})}^2.$$

- Similar to $\mathcal{F}_{\text{KL}}$, $\mathcal{F}_{\chi^2}(\cdot) = \chi^2(\cdot, \mathbb{Q})$ flow enjoys convergence for Gaussian-like $\mathbb{Q}$ [OT11].
- An **interpolation** between MMD^2 and χ^2 ?

DrMMD: An interpolation of MMD and χ^2

Definition (De-regularized Maximum Mean Discrepancy (DrMMD))

Suppose $\mathbb{P} \ll \mathbb{Q}$ where $\mathbb{P}, \mathbb{Q} \in \mathcal{P}_2(\mathbb{R}^d)$. Then the (de)-regularized maximum mean discrepancy (DrMMD) between $\mathbb{P}, \mathbb{Q}$ is defined as, for $\lambda > 0$,

$$\text{DrMMD}(\mathbb{P}, \mathbb{Q}) = (1 + \lambda) \left\| \left((T_{\mathbb{Q}} + \lambda \text{Id})^{-1} T_{\mathbb{Q}} \right)^{\frac{1}{2}} \left(\frac{d\mathbb{P}}{d\mathbb{Q}} - 1 \right) \right\|_{L_2(\mathbb{Q})}^2.$$

Proposition 2 (Interpolation of MMD^2 and χ^2)

$$\lim_{\lambda \rightarrow 0} \text{DrMMD}(\mathbb{P}, \mathbb{Q}) = \chi^2(\mathbb{P}, \mathbb{Q}), \quad \lim_{\lambda \rightarrow \infty} \text{DrMMD}(\mathbb{P}, \mathbb{Q}) = \text{MMD}^2(\mathbb{P}, \mathbb{Q}).$$

- Similar ideas in kernel hypothesis testing [MRW⁺99, HBM07, HKSL24].
- Interpolation between MMD and general f-divergences [GAG21, SNRS26].

DrMMD gradient flow

Definition ($\mathcal{F}_{\text{DrMMD}}$ gradient flow)

The Wasserstein gradient flow of $\mathcal{F}_{\text{DrMMD}}(\cdot) = \text{DrMMD}(\cdot, \mathbb{Q})$ is $(\mathbb{P}_t)_{t \geq 0}$ with

$$\partial_t \mathbb{P}_t + \nabla \cdot (\mathbb{P}_t (1 + \lambda) \nabla h_{\mathbb{P}_t, \mathbb{Q}}) = 0.$$

where $h_{\mathbb{P}, \mathbb{Q}} = (T_{\mathbb{Q}} + \lambda \text{Id})^{-1} T_{\mathbb{Q}} \left(\frac{d\mathbb{P}}{d\mathbb{Q}} - 1 \right)$.

- Implementation: $h_{\mathbb{P}, \mathbb{Q}} = (T_{\mathbb{Q}} + \lambda \text{Id})^{-1} (m_{\mathbb{P}} - m_{\mathbb{Q}})$ with $m_{\mathbb{P}}, m_{\mathbb{Q}}$ the mean embeddings.
- MMD goes to grad school, becomes DrMMD, then drives a new Wasserstein gradient flow.



DrMMD: Convergence of $\mathcal{F}_{\chi^2}$ flow

Definition (Poincaré inequality)

We say that $\mathbb{Q}$ satisfies a **Poincaré inequality** with constant C_P if for all $f, \nabla f \in L_2(\mathbb{Q})$,

$$\text{Var}_{\mathbb{Q}}[f] \leq C_P \mathbb{E}_{\mathbb{Q}} [\|\nabla f\|^2] .$$

- By “Gaussian-like,” we mean that $\mathbb{Q}$ satisfies a Poincaré inequality, which in particular implies **sub-exponential tails** [BGL⁺14].
- For example, if $\mathbb{Q} = \mathcal{N}(0, a\text{Id})$, then $C_P = a$.

Proposition 3 (Convergence of $\mathcal{F}_{\chi^2}$ flow)

Suppose that $\mathbb{Q}$ satisfies the Poincaré inequality. Let $(\mathbb{P}_t)_{t \geq 0}$ follow the $\mathcal{F}_{\chi^2}$ flow. Then, for any $T \geq 0$, $\text{KL}(\mathbb{P}_T, \mathbb{Q}) \leq \exp(-2C_P^{-1}T) \text{KL}(\mathbb{P}_0, \mathbb{Q})$.

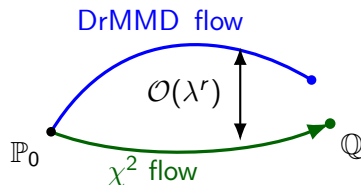
DrMMD: Convergence of DrMMD gradient flow

Theorem 2 (Convergence of DrMMD gradient flow)

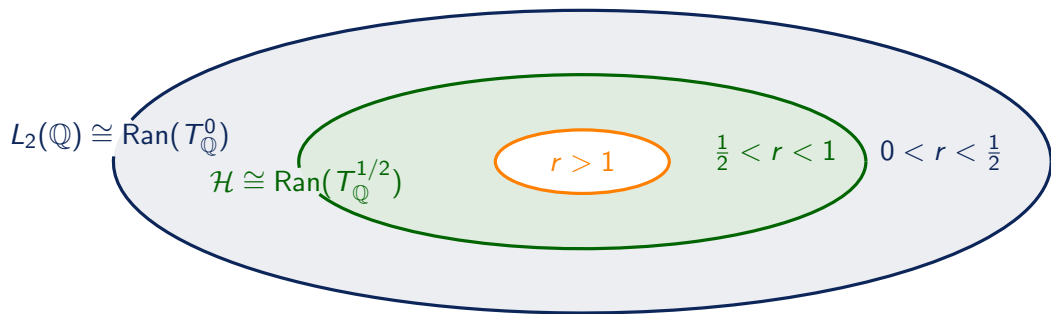
Let $(\mathbb{P}_t)_{t \geq 0}$ be the $\mathcal{F}_{\text{DrMMD}}$ flow. Suppose the target distribution $\mathbb{Q}$ satisfies a Poincaré inequality with constant C_P . Suppose that $\frac{d\mathbb{P}_t}{d\mathbb{Q}} - 1 \in \text{Ran}(\mathcal{T}_{\mathbb{Q}}^r)$ with $r > 0$. Then for any $T \geq 0$,

$$\text{KL}(\mathbb{P}_T, \mathbb{Q}) \leq \exp\left(-\frac{2(1+\lambda)}{C_P} T\right) \text{KL}(\mathbb{P}_0, \mathbb{Q}) + \mathcal{O}(\lambda^r C_P).$$

- Larger r means more regularity of the trajectory and thus smaller bias.
- A smaller Poincaré constant C_P means faster convergence.
- For continuous-time DrMMD flow, we want $\lambda \rightarrow 0$ for ‘convexity’; this is not the case for discrete time.



DrMMD: Interpolation space $\text{Ran}(T_{\mathbb{Q}}^r)$



- Large r indicates larger smoothness.

- DrMMD gradient flow (continuity equation)

$$\partial_t \mathbb{P}_t + \nabla \cdot (\mathbb{P}_t(1 + \lambda) \nabla h_{\mathbb{P}_t, \mathbb{Q}}) = 0$$

- DrMMD gradient descent: for step size $\gamma > 0$,

$$\mathbb{P}_{n+1} = (\text{Id} + \gamma(1 + \lambda) \nabla h_{\mathbb{P}_n, \mathbb{Q}})_{\#} \mathbb{P}_n.$$

DrMMD: Convergence of DrMMD gradient descent

Proposition 4 (Descent lemma of DrMMD gradient descent)

Let $(\mathbb{P}_n)_{n \in \mathbb{N}}$ be the Wasserstein gradient descent of $\mathcal{F}_{\text{DrMMD}}$. Suppose $\mathbb{Q} \propto \exp(-V)$ with $\mathbf{H}V \preceq L\text{Id}$. Suppose the step size γ is small enough.

$$\text{KL}(\mathbb{P}_{n+1}, \mathbb{Q}) - \text{KL}(\mathbb{P}_n, \mathbb{Q}) \leq -\frac{2}{C_P} \chi^2(\mathbb{P}_n, \mathbb{Q}) \gamma + \underbrace{\gamma \lambda^r}_{\text{Approximation error}} + \underbrace{\gamma^2 L \chi^2(\mathbb{P}_n, \mathbb{Q}) \frac{1}{\lambda}}_{\text{Discretization error}}$$

- A **trade-off** on λ between approximation error and time-discretization error.
- Optimal choice of **adaptive** λ_n at each iterate n :

$$\lambda_n \propto \chi^2(\mathbb{P}_n, \mathbb{Q})^{\frac{1}{r+1}}$$

- At the start, a **larger** λ gives stronger regularization; closer to convergence, a **smaller** λ operates in the χ^2 regime and better captures the difference.

DrMMD: Convergence of DrMMD gradient descent

Theorem 3 (Convergence of DrMMD gradient descent)

Let $(\mathbb{P}_n)_{n \in \mathbb{N}}$ be the Wasserstein gradient descent of $\mathcal{F}_{\text{DrMMD}}$. Suppose all conditions from the descent lemma hold. Then, for any $n_{\max} \in \mathbb{N}$,

$$\text{KL}(\mathbb{P}_{n_{\max}}, \mathbb{Q}) \leq \exp\left(-\frac{2n_{\max}\gamma}{C_P}\right) \text{KL}(\mathbb{P}_0, \mathbb{Q}) + \mathcal{O}(\gamma^{\frac{r}{r+1}} C_P L^{\frac{r}{r+1}}).$$

- To reach error $\text{KL}(\mathbb{P}_{n_{\max}}, \mathbb{Q}) \leq \delta$, it takes $\mathcal{O}((\frac{1}{\delta})^{\frac{r+1}{r}} \log \frac{1}{\delta})$ iterations.
- By comparison, for Langevin Monte Carlo, it takes $\mathcal{O}(\frac{1}{\delta} \log \frac{1}{\delta})$ [CEL⁺24].
- DrMMD gradient descent takes more iterations due to the additional approximation error $\mathcal{O}(\lambda^r)$, but it **operates without the knowledge of potential V** and only requires $\Sigma_{\mathbb{Q}} = \int k(x, \cdot) \otimes k(x, \cdot) d\mathbb{Q}$ and the embedding $\int k(x, \cdot) d\mathbb{Q}$.

SrMMD: motivation

- Recall DrMMD.

$$\text{DrMMD}(\mathbb{P}, \mathbb{Q}) = \sup_{f \in \mathcal{H}: \|f\|_{L_2(\mathbb{P})}^2 + \lambda \|f\|_{\mathcal{H}}^2 \leq 1} \int f \, d(\mathbb{P} - \mathbb{Q}).$$

Definition (Sobolev-regularized Maximum Mean Discrepancy (SrMMD))

$$\text{SrMMD}(\mathbb{P}, \mathbb{Q}) = \sup_{f \in \mathcal{H}: \|\nabla f\|_{L_2^d(\mathbb{P})}^2 + \lambda \|f\|_{\mathcal{H}}^2 \leq 1} \int f \, d(\mathbb{P} - \mathbb{Q}).$$

- This was first proposed in [MSR19].

SrMMD: motivation

- Define $S_{\mathbb{P}} : L_2(\mathbb{P}) \rightarrow L_2(\mathbb{P})$ as the **gradient covariance** operator:

$$S_{\mathbb{P}} f(\cdot) = \int \nabla f(x)^{\top} \nabla_1 k(x, \cdot) d\mathbb{P}(x).$$

- Define $m_{\mathbb{P}} = \int k(x, \cdot) d\mathbb{P}(x)$ as the kernel mean embedding.

Definition (SrMMD gradient flow)

The SrMMD flow is $(\mathbb{P}_t)_{t \geq 0}$ that satisfies

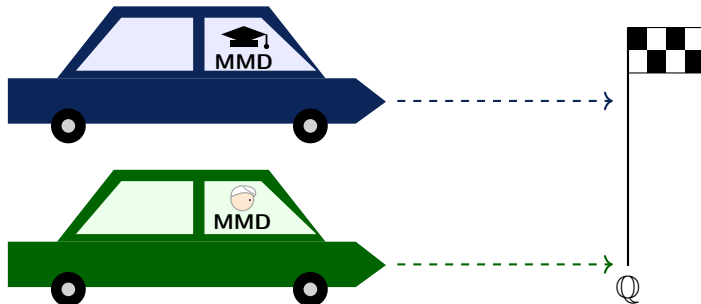
$$\partial_t \mathbb{P}_t + \nabla \cdot (\mathbb{P}_t (1 + \lambda) \nabla f_{\mathbb{P}_t, \mathbb{Q}}) = 0.$$

where $f_{\mathbb{P}, \mathbb{Q}} = (\mathbf{S}_{\mathbb{P}} + \lambda \text{Id})^{-1} (m_{\mathbb{P}} - m_{\mathbb{Q}})$.

- For comparison, $f_{\mathbb{P}, \mathbb{Q}} = (\mathbf{T}_{\mathbb{Q}} + \lambda \text{Id})^{-1} (m_{\mathbb{P}} - m_{\mathbb{Q}})$ for DrMMD flow.

SrMMD: SrMMD gradient flow

- MMD gets old: becomes SrMMD, then drives a new Wasserstein gradient flow.



SrMMD: Convergence

Theorem 4 (Convergence of SrMMD flow)

Let $(\mathbb{P}_t)_{t \geq 0}$ be the SrMMD flow. Suppose that there exists $0 < r \leq 1$ such that $m_{\mathbb{P}_t} - m_{\mathbb{Q}} \in \text{Ran}(S_{\mathbb{P}_t}^{r/2})$ for all t . Then, for any $T \geq 0$,

$$\text{MMD}^2(\mathbb{P}_T, \mathbb{Q}) \leq \exp(-2T) \text{MMD}^2(\mathbb{P}_0, \mathbb{Q}) + \mathcal{O}(\lambda^r).$$

Theorem 5 (Convergence of SrMMD descent)

Let $(\mathbb{P}_n)_{n \in \mathbb{N}}$ be the SrMMD descent. Suppose that there exists $0 < r \leq 1$ such that $m_{\mathbb{P}_n} - m_{\mathbb{Q}} \in \text{Ran}(S_{\mathbb{P}_n}^{r/2})$ for all n . For step size $\gamma > 0$, take $\lambda \propto d\gamma$. Then, for any $n_{\max} \geq 0$,

$$\text{MMD}^2(\mathbb{P}_{n_{\max}}, \mathbb{Q}) \leq \exp(-2n_{\max}\gamma) \text{MMD}^2(\mathbb{P}_0, \mathbb{Q}) + \mathcal{O}(\gamma^r).$$

- On a torus, $\text{Ran}(S_{\mathbb{P}}^{r/2}) \cong H^{s(r+1)-r}$, where s is the kernel order.
- $\lambda \propto d\gamma$ balances **regularity error** $\mathcal{O}(\lambda^r \gamma)$ and **discretization error** $\mathcal{O}(\lambda^{r-1} \gamma^2)$.

SrMMD: Iteration complexity

- Number of iterations sufficient to reach δ -closeness to $\mathbb{Q}$.

Algorithm	Metric	Iteration complexity
LMC [CEL+24, VW19]	KL / χ^2	$\tilde{\mathcal{O}}(\delta^{-1}\beta)$
SVGD [BBG25]	KSD ²	$\mathcal{O}(\delta^{-1})$
R-SVGD [HBSL24]	Fisher	$\mathcal{O}(\delta^{-1})$
DrMMD	KL	$\tilde{\mathcal{O}}(\delta^{-(r+1)/r}\beta)$
SrMMD	MMD ² / KSD ²	$\tilde{\mathcal{O}}(\delta^{-1/r})$

Table: Here, β^{-1} denotes the LSI constant of $\mathbb{Q}$, and r denotes the regularity parameter of the flow.

- Different metrics: χ^2 stronger than KL and both are stronger than MMD/KSD [GM17, KBGM25].
- The SrMMD rate is **independent** of the LSI constant β^{-1} .

In the following, we turn to applications

1) Minimum MMD estimation 2) Generative models



François-Xavier
Briol



Sophia
Kang



Louis
Sharrock



Arthur
Gretton

Application: Statistical inference

- $\mathbb{P}_\theta$ is the model; and $\mathbb{Q}$ is the data distribution.
- Maximum Likelihood estimator: $\arg \min_{\theta \in \Theta} \text{KL}(\mathbb{Q}, \mathbb{P}_\theta)$.
- Minimum MMD estimator [BBDG19, CAA22, ACADF23a, AG25]

$$\begin{aligned}\theta_{\text{MMD}} &= \arg \min_{\theta \in \Theta} \text{MMD}^2(\mathbb{P}_\theta, \mathbb{Q}) \\ &= \arg \min_{\theta \in \Theta} \left\{ \mathbb{E}_{x, x' \sim \mathbb{P}_\theta} [k(x, x')] - 2\mathbb{E}_{x \sim \mathbb{P}_\theta, y \sim \mathbb{Q}} [k(x, y)] + \text{const} \right\}.\end{aligned}$$

- Applications: [KGBF25, SKA⁺23, BFM25, FAZL23, AG24, AFP26, DOD25, DKDB22, DD23, CDB25, SK25, ACADF23b, CAN25, KPLB26].

Advantage 1: Likelihood-free inference?

✓ Only requires samples from $\mathbb{P}_\theta$ and $\mathbb{Q}$.

Advantage 2: Robustness to outliers?

✓ Bounded effects under bounded kernels.

Application: Statistical inference

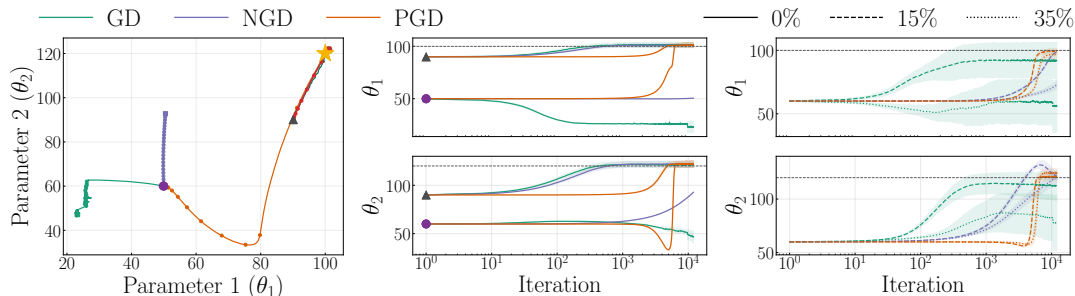
Question: How can we reliably find θ_{MMD} ?

1. MMD gradient flow $\mathbb{P}_{t+1} = (\text{Id} - \gamma \nabla f_{\mathbb{P}_t, \mathbb{Q}})_{\#} \mathbb{P}_t$.
2. Consider the model family: $\mathbb{P}_{\theta} = (T_{\theta})_{\#} \rho$: $\mathbb{P}_{\theta_{t+1}} = (T_{\theta_t - \gamma \Delta \theta_t})_{\#} \rho$.
3. Choose $\mathbb{P}_{\theta_{t+1}}$ closest to the MMD flow iterate $\mathbb{P}_{t+1}$.

$$\Delta \theta_t^{\text{PGD}} = \underbrace{\left\{ \mathbb{E}_{z \sim \rho} [\mathbf{J}_{\theta} G_{\theta_t}(z)^{\top} \mathbf{J}_{\theta} G_{\theta_t}(z)] \right\}^{-1}}_{\text{preconditioner}} \underbrace{\nabla_{\theta} \text{MMD}^2(\mathbb{P}_{\theta_t}, \mathbb{Q})}_{\text{standard gradient}}.$$

- Convergence: $\lim_{t \rightarrow \infty} \text{MMD}(\mathbb{P}_{\theta_t}, \mathbb{Q}) = \min_{\theta} \text{MMD}(\mathbb{P}_{\theta}, \mathbb{Q})$.

Experiment: Lotka–Volterra inference



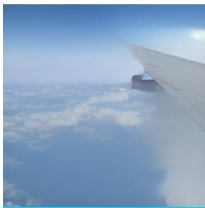
- We compare gradient descent (GD), natural gradient descent (NGD) [BBDG19], and preconditioned gradient descent (PGD, ours).
- First row: Close initialization. Second row: Far initialization.

Experiment: composite goodness-of-fit testing [KGBF25]

Method	Type-I error ($\downarrow$)	Time	Power ($\uparrow$)	Time
PGD ($\mathcal{I} = 1$)	0.050 ± 0.018	8.6s	0.829 ± 0.032	3.6s
GD ($\mathcal{I} = 400$)	0.107 ± 0.026	10.4s	0.600 ± 0.041	4.9s
GD ($\mathcal{I} = 1$)	0.071 ± 0.022	2.6s	0.279 ± 0.038	1.1s

Table: Composite goodness-of-fit test. $\mathcal{I}$ represents the number of restarts.

Application: Generative models (WIP)



wing



hammerhead shark



Irish terrier



coyote



schooner



water buffalo



manhole cover



fire engine



vending machine



dining table

- ResNet-style encoder pre-trained with masked autoencoding (MAE) [DLL⁺26].

Conclusions

- Minimization of MMD
 - Interpolation towards χ^2 :

$$\text{DrMMD}(\mathbb{P}, \mathbb{Q}) = \sup_{f \in \mathcal{H}: \|f\|_{L_2(\mathbb{P})}^2 + \lambda \|f\|_{\mathcal{H}}^2 \leq 1} \int f \, d(\mathbb{P} - \mathbb{Q})$$

- Gradient penalty:

$$\text{SrMMD}(\mathbb{P}, \mathbb{Q}) = \sup_{f \in \mathcal{H}: \|\nabla f\|_{L_2^d(\mathbb{P})}^2 + \lambda \|f\|_{\mathcal{H}}^2 \leq 1} \int f \, d(\mathbb{P} - \mathbb{Q})$$

- Application I: Minimum MMD inference.
- Application II: One-step generative models.

About Me



- Zonghao Chen
- 4th year PhD Student at University College London (UCL)
 - Foundational AI Centre
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- Graduated from Tsinghua University in 2022
 - Department of EE
- Kernel (nonparametric) methods, causal inference, statistical learning theory

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